1. This question consists of 6 sub-questions. Each item can be answered in a few words or a short paragraph with use of a diagram and/or expression where helpful. Please answer all sub-questions and simplify expressions where possible stating any assumptions made.

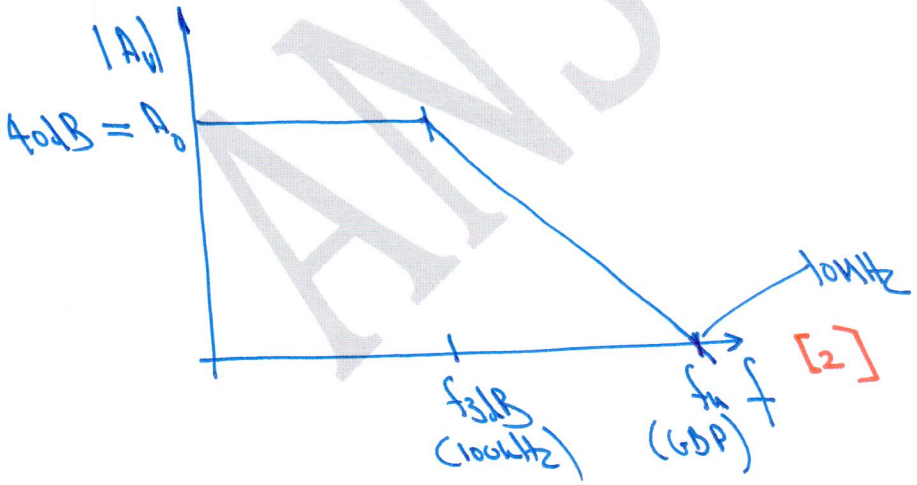
a) Explain why it may be preferable to implement a single transistor amplifier using a PMOS device instead of using a NMOS device.

[5]

This is preferred for single amplifier configurations that do NOT have the source terminal of MOSFET connected to a common node, i.e. Power Supply. [1] for example, CS with degen, diff pair, or source follower [1] This is because of body effect impacting NMOS devices (they all share same substrate = body $\therefore V_{SB} \neq 0$) [1] with PMOS amplifiers they can have own N-well $\therefore V_{SB} = 0$ is possible. With NMOS possible to do [1] this only if technology provides a triple-well option. [1]

b) The frequency response of an operational amplifier can be described in terms of its -3dB bandwidth, unity gain bandwidth and open loop gain. Given a unity gain bandwidth of 10MHz and open loop gain of 40dB, what is expected to be the gain at a frequency of 25kHz?

[5]



GBP = f_u = 10MHz

$= A_0 \times f_{3dB} \rightarrow f_{3dB} = \frac{10MHz}{100} = 100kHz$ [1]

[1]

$\therefore$ @ $f = 25kHz$, $A_v = 40dB$ [1]

ANSWERS

c) Derive expressions (by inspection) for the voltage gains of the amplifier circuits shown below in Fig. 1.1. Include the effect of a finite output resistance.

[10]

(i)

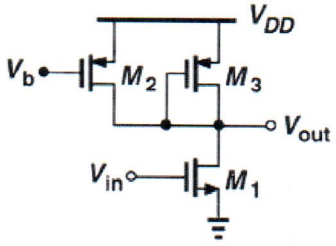


Fig. 1.1(a)

(ii)

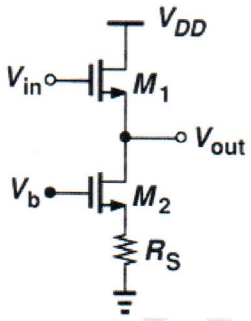


Fig. 1.1(b)

M_1 = CS amp, M_2/M_3 act as load [1]

$A_v = -g_{m1} \times R_{out}$ [1]

$R_{out} = r_{o1} \| r_{o2} \| r_{o3} \| 1/g_{m3}$ [1]

(Assuming $1/g_{m3} \ll r_{o3}$) [1]

$\therefore A_v \approx g_{m1}(r_{o1} \| r_{o2} \| 1/g_{m3})$ [1]

M_1 = CD/SF amp, M_2/R_S S degen load [1]

$A_v = \dfrac{R_D \| r_o}{R_D \| r_o + 1/g_m}$ [1]

$R_D = g_{m2} r_{o2} R_S + r_{o2} + R_S$

(assuming $g_m r_o \gg 1$) [1]

$A_v \approx \dfrac{g_{m2} r_{o2} R_S \| r_{o1}}{g_{m2} r_{o2} R_S \| r_{o1} + 1/g_{m1}}$ [1]

d) The differentiator shown below in Fig. 1.2 is used to amplify a sinusoidal input at a frequency of 1 MHz by a factor of 5. If A0=∞, determine the value of R_1C_1.

[5]

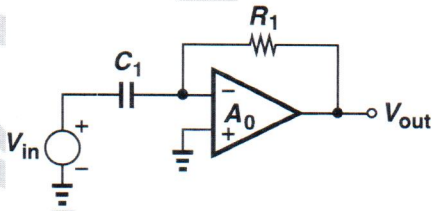


Fig. 1.2

$A_0 = \infty$, $A_v = \dfrac{V_{out}}{V_{in}} \Rightarrow \dfrac{V_{out}}{R_1} = -V_{in} \cdot \omega \cdot C_1$ [1]

$\left|\dfrac{V_{out}}{V_{in}}\right| = \omega R_1 C_1 = 5$ [1]

$R_1C_1 = \dfrac{5}{\omega}$, $\omega = 2\pi f$ [1]

$R_1C_1 = \dfrac{5}{2\pi \times 10^6} = 7.96 \times 10^{-5}$ (s)

[1] [1]

e) Assuming ~~perfect symmetry and~~ $V_A<\infty$, compute the differential voltage gain of each stage shown below in Fig. 1.3. State any assumptions.

[5]

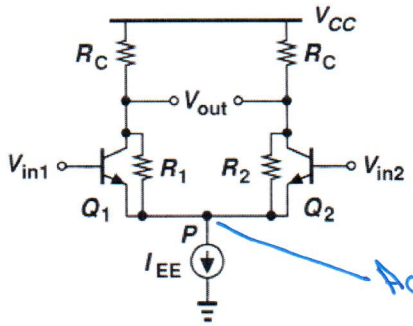
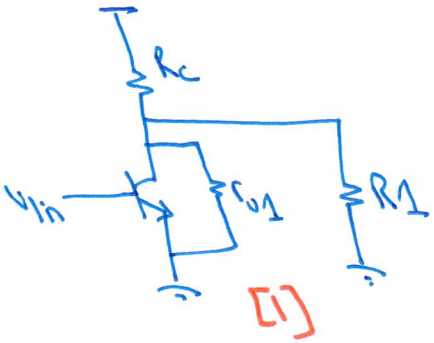


Fig. 1.3

Half circuit approximation (perfect symmetry) [1]



[1]

This is a CE amplifier [1]

$A_v = -g_{m1}(R_C \| R_1 \| r_{o1})$ [2]

f) An adventurous student designs the following circuit shown in Fig. 1.4. Unfortunately, the amplifier does not operate well.

(i) Prove by inspection that the feedback is positive [5]

(ii) Breaking the loop at the gate of M2, determine the loop gain and prove that the feedback is positive.

[5]

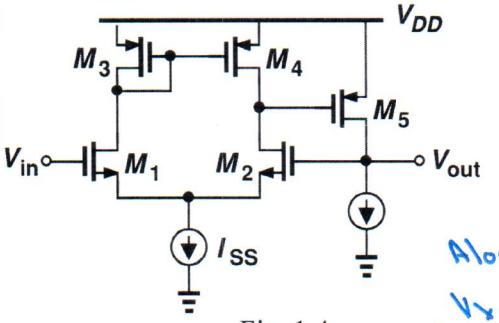
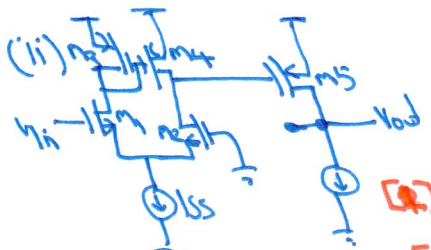


Fig. 1.4



[2]

$A_{loop} = \frac{v_{out}}{v_{in}} = \frac{v_x}{v_{in}} \cdot \frac{v_{out}}{v_x}$ [1] from SS $r_x = r_{o2} \| r_{o4}$

$v_x = +g_{m1} v_{in} (r_{o2} \| r_{o4})$

2nd stage: $v_{out} = -g_{m5} r_{o5} v_x$ [1] $\therefore A_{loop} = g_{m1} g_{m5} (r_{o2} \| r_{o4}) r_{o5}$ [1]

Since loop gain -ve $\therefore f_b$ = +ve [1]

[2]

(i) $\uparrow v_{in}, \downarrow V_{GS}, \uparrow I_{D1}, \uparrow V_{GS}, \downarrow V_{out}$ (V_{G2}) $\therefore V_{GS} \uparrow$

- v_{in} is in phase with V_{GS} [1] $\therefore$ +ve feedback.
- M_5 is CS $\therefore$ 180° out of phase [1]
- V_{out} is I/P to diff. pair → out of phase by 180° [1]

ANSWERS

2. The circuit shown below in Fig. 3.1. is a two stage amplifier, specifically a common source stage followed by a source follower stage.

Assume all devices are in saturation and $\lambda>0$ (i.e. $R_{out}<\infty$).

Where required, use the following values:

$\mu_n C_{ox}=200\mu A/V^2$	$V_{TH,n}=0.4V$	$\lambda_n=0.1V^{-1}$
$C_{GS}=(2/3)WLC_{ox}$	$C_{ox}=12fF/\mu m^2$	$C_{DB}\approx 0$
$C_{GD}=C_0W$	$C_0=0.2fF/\mu m$	$C_{SB}\approx 0$

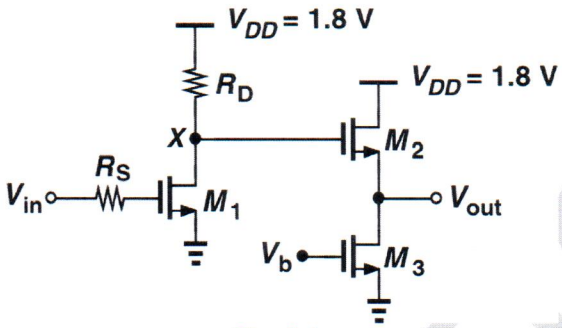


Fig. 3.1.

a) Determine an expression for the low frequency voltage gains of the two stages, i.e. (V_x/V_{in}) and (V_{out}/V_x), and hence the overall low frequency voltage gain (V_{out}/V_{in}).

[5]

$\frac{V_x}{V_{in}} = -g_{m1}(r_{o1}\|R_D)$ [1] CS amp. [1]

$\frac{V_{out}}{V_x} = \frac{r_{o3}\|r_{o2}}{r_{o3}\|r_{o2}+1/g_{m2}}$ [1] CD amp [1]

$A_V = \frac{-g_{m1}(r_{o1}\|R_D)(r_{o2}\|r_{o3})}{r_{o2}\|r_{o3}+1/g_{m2}}$ [1]

$R_D = 1k\Omega$

b) Evaluate the low frequency voltage gains, given $(W/L)_1=(W/L)_2=(W/L)_3=10\mu m/0.18\mu m$, and $I_{D1}=I_{D2}=I_{D3}=1mA$.

[5]

$g_{m1} = \sqrt{2I_D(W/L)_1\mu_n C_{ox}} = \sqrt{2(200\mu)\cdot\frac{10}{0.18}\cdot 1mA} = 4.71mS$

$g_{m2} = \sqrt{2I_D(W/L)_1\mu_n C_{ox}} = 4.71mS$ [1]

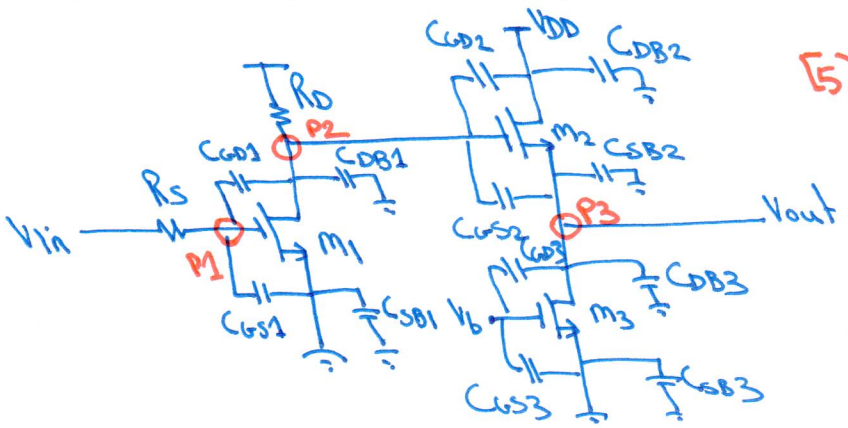
$r_{o1} = r_{o2} = r_{o3} = \frac{1}{\lambda I_D} = \frac{1}{0.1\times 1mA} = 10k\Omega$ [1]

$A_V = -3 = \frac{V_{out}}{V_{in}}$ [1]

$\therefore \frac{V_x}{V_{in}} = -4.28$ [1] $\quad \frac{V_{out}}{V_x} = 0.702$ [1]

c) Redraw the circuit above to: (i) identify the 3 nodes that are associated with poles and (ii) include all the parasitic device capacitances (within $M_{1\text{-}3}$).

[5]

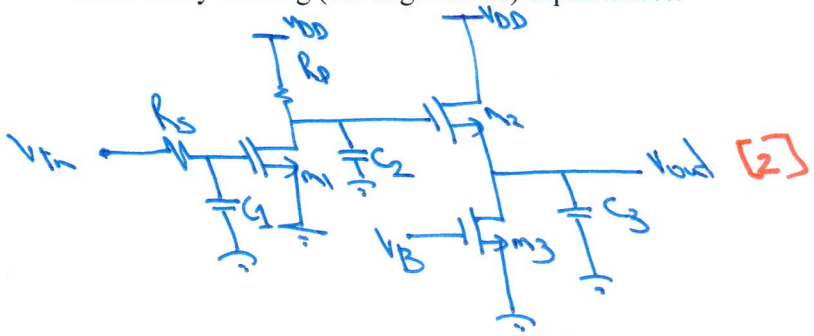


[5] -½ per error or omission

P1, P2, P3 nodes associated with poles.

ANSWERS

d) Determine expressions for the capacitances (to ground) at each of the nodes identified above by lumping together the parasitic capacitances appearing at each node. Use Miller's theorem to resolve any floating (i.e. ungrounded) capacitances. [5]



[2]

$C_1 = C_{GS1} + C_{GD1}(1+|A_{V1}|) = C_{GS1} + 5.3 \times C_{GD1}$ [1]

$C_2 = C_{DD1} + C_{DD2} + C_{GS2}(1-A_{V2}) + C_{GD1}\left(1+\frac{1}{A_{V1}}\right)$

$= 0.3 \times C_{GS2} + 1.24 \times C_{GD1}$ [1]

$C_3 = C_{DD3} + C_{SB2} + C_{GS2}\left(1-\frac{1}{A_{V2}}\right) + C_{GD3}$

$= C_{GD3} - 0.43 C_{GS2}$ [1]

e) Determine expressions for the ~~three~~ two pole frequencies (f_{p1}, f_{p2} ~~and f_{p3}~~) and evaluate. given $R_S = 10k\Omega$. excluding the output pole [5]

$f_{p1} = \frac{1}{2\pi R_S \cdot C_1}$ [1] $\quad f_{p2} = \frac{1}{2\pi (R_D \| r_{O1}) C_2}$ [1]

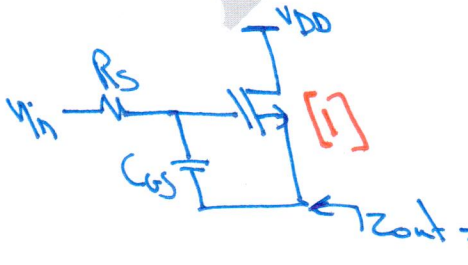
$C_1 = \frac{2}{3}(10 \times 0.18) \times 12f + 0.2(10) \times 5.3f = 25ff$

$C_2 = 0.3\left(\frac{2}{3}\right)(10 \times 0.18) \times 12ff + 1.24 \times 0.2(10) = 6.8ff$ [1]

$f_{p1} = \frac{1}{2\pi (10k) 25ff} = 636.6\,MHz$ [1]

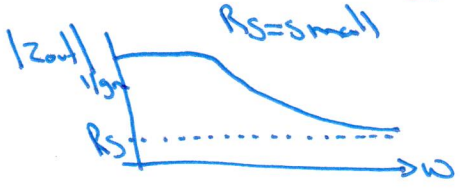
$f_{p2} = \frac{1}{2\pi (1k \| 10k) 6.8ff} = 2.57\,GHz$ [1]

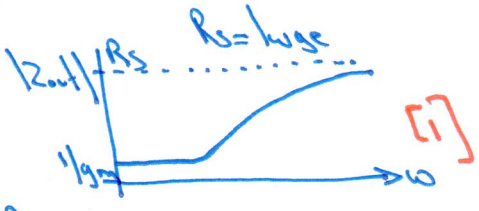
f) Comment on the high frequency output impedance of the circuit. [5]



[1]

$Z_{out} = \frac{R_S C_{GS} s + 1}{C_{GS} s + g_m}$ [1]





[1]

Complex output impedance that can be inductive depending on relative impedance between 1/gm and RS (driving O/P impedance) [2]

ANSWERS

3. The circuit shown below in Fig. 2.1. is called a *Operational Transconductance Amplifier (OTA).*

The circuit operates as follows: Device M9 provides the bias current (sink) to the differential pair M_1/M_2. The two tail currents (I_{D1} and I_{D2}) are then copied by the active current mirrors M_3/M_4, M_5/M_6 and M_7/M_8 such as to subtract current I_1 from I_2 at the output node. This output current (i.e. I_2 - I_1) is typically delivered to an external load. Mirrors M_5/M_6 and M_7/M_8 can be scaled 1:K to provide additional current gain.

Assume all devices are in saturation and $\lambda>0$ (i.e. $R_{out}<\infty$).

Where required, use the following values: $\mu_nC_{ox}=200\mu A/V^2$ $V_{TH,n}=0.4V$ $\lambda_n=0.1V^{-1}$
$\mu_pC_{ox}=100\mu A/V^2$ $V_{TH,p}=-0.5V$ $\lambda_p=0.2V^{-1}$

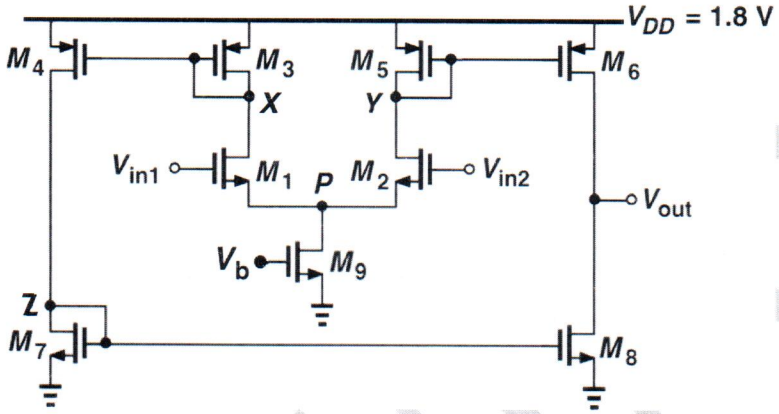


Fig. 2.1.

a) This OTA can be used as a voltage amplifier, provided that the load resistance is large. Determine an expression for the voltage gain of the OTA, i.e. $A_v=V_{out}/(V_{in2}-V_{in1})$ given that $(W/L)_6=K(W/L)_5$ and $(W/L)_8=K(W/L)_7$.

[5]

Assuming perfect symmetry. (eg. $g_{m1}=g_{m2}$) [1]

$A_v = g_{m1} \times K \times (r_{o6} \| r_{o8})$ [1]

– correctly identifying current gain [1]
– correctly " transconductance amp [1]
– " " o/p impedance. [1]

b) Given that $I_{D9}=100\mu A$, $(W/L)_{1,2}=50/3$, $(W/L)_{3,4,5,7,9}=3/3$, K=5, evaluate the Voltage Gain and Power Consumption.

[5]

$g_{m1} = \sqrt{2 \times \mu_n C_{ox} \times W/L \times I_D} = \sqrt{2(200\mu)\frac{50}{3} \cdot 100\mu A} = 816\mu S$ [1]

$r_{o6} = \frac{1}{\lambda_p I_D} = \frac{1}{0.2(\frac{100\mu}{100})5} = 10k\Omega$ [1]

$r_{o8} = \frac{1}{\lambda_n I_D} = \frac{1}{0.1(100\mu)5} = 20k\Omega$

$A_v = 816\mu S \times 5 \times 6.6k\Omega = 27.2$ [1]

$= 28.7 dB$ [1]

Power $= 1.8 \times 3 \times 100\mu A + 5 \times 100\mu A$
$= 1.44mW.$ [1]

c) How would any variation in the bias current impact the voltage gain? In practice why is it challenging to make stable, well-defined bias currents on-chip?

[5]

Any variation in bias current would effect g_m by $\sqrt{\Delta I_D}$ but also the r_o of the devices M_6 and M_8 inversely proportional [1]

$\therefore \Delta A_V = \frac{\sqrt{\Delta I_D}}{\Delta I_D} = \frac{1}{\sqrt{\Delta I_D}}$ [1] so a 10% increase in bias current → 4.65% increase in A_0. [1]

Generally it is challenging to implement precise and stable current reference on chip due to process variation, PSRR (noise) and temperature dependance. [2]

d) Comment on this amplifiers expected high frequency performance compared to a ~~single~~ two stage op-amp, i.e. differential pair with active current mirror load.

[5]

The OTA has high impedance nodes only at I/P and O/P. [1] All internal nodes are low impedance, [1] whereas standard op-amp has high impedance internal node also [1] ∴ Expect this to have better high freq. performance. [1]

Also OTA can be load compensated ← desirable. while 2-stage miller compensated. [1]

e) Describe how the circuit can be modified to:

(i) Increase its voltage gain; [5]

Cascode output stage → increase R_{out}. [2]

Increase K → current amplification [2]

Increase $g_{m1} = g_{m2}$ → W/L [1]

Not bias current

(ii) Use as a typical operational amplifier. [5]

→ need drive strength, lower R_{out} [2]

Add a 2nd stage → can be a CS stage providing additional gain but will then need compensation capacitor (note change of I/P terminals). [3]

Another option is a push-pull O/P stage (i.e. CD amplifiers) or

→ needs biasing circuits

ANSWERS